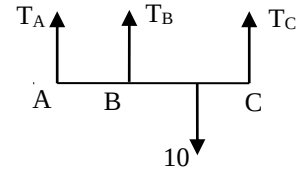


THE UNIVERSITY OF BIRMINGHAM
DEPARTMENT OF CIVIL ENGINEERING

Mechanics 1

Tutorial Sheet - 3

1. Resolve vertically:
 $T_A + T_B + T_C - 10 = 0$
 $T_A + T_B + T_C = 10$ [1]

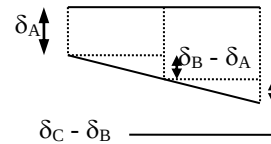


Taking moments about point A (+ve clockwise)
 $-2T_B + 30 - 5T_C = 0$
 $2T_B + 5T_C = 30$ [2]

From theory of elasticity: $\delta_A = \frac{T_A L}{EA}$ $\delta_B = \frac{T_B L}{EA}$
 $\delta_C = \frac{T_C L}{EA}$

Displacements are as follows:

By similar triangles:



$$\frac{\delta_A - \delta_B}{2} = \frac{\delta_B - \delta_C}{3}$$

$$\Rightarrow 3\delta_B - 3\delta_A = 2\delta_C - 2\delta_B$$

$$-3\delta_A + 5\delta_B - 2\delta_C = 0$$

$$-3\frac{T_A L}{EA} + 5\frac{T_B L}{EA} - 2\frac{T_C L}{EA} = 0$$

$$-3T_A + 5T_B - 2T_C = 0$$
 [3]

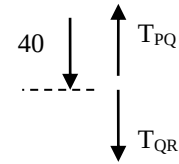
$$3 \times [1] + [3] \quad 8T_B + T_C = 30 \quad [4]$$

$$4 \times [2] - [4] \quad 19T_C = 90 \Rightarrow T_C = 4.737 \text{ kN}$$

Subs in [2] $2T_B + 5T_C = 30$ $T_B = 3.158 \text{ kN}$
 $T_A = 2.105 \text{ kN}$

2 Assume each part is in tension

$$\begin{aligned} \text{Resolve vertically: } T_{PQ} - T_{QR} - 40 &= 0 \\ T_{PQ} - T_{QR} &= 40 \quad [1] \end{aligned}$$



Using theory of elasticity:

$$\delta_{PQ} = \frac{3T_{PQ}}{EA} \quad \delta_{QR} = \frac{4.5T_{QR}}{EA}$$

Using compatibility: $\delta_{PQ} + \delta_{QR} = 0$

$$\begin{aligned} \frac{3T_{PQ}}{EA} + \frac{4.5T_{QR}}{EA} &= 0 \\ T_{PQ} &= -1.5 T_{QR} \quad [2] \end{aligned}$$

Solving [1] and [2] gives:

$$T_{QR} = -16 \text{ kN (compression)}$$

$$T_{PQ} = 24 \text{ kN (tension)}$$

3 Consider temperature rise:

$$\text{Material 1, } \epsilon_1 = \alpha_1 DT = 1.1 \times 10^{-5} \times 20 = 2.2 \times 10^{-4}$$

$$\text{Material 12 } \epsilon_2 = \alpha_2 DT = 7.0 \times 10^{-5} \times 20 = 1.4 \times 10^{-3}$$

$$\text{Resolving forces: } P_1 + P_2 = 0 \Rightarrow P_2 = -P_1$$

Using compatibility: $\epsilon_1 = \epsilon_2$

$$\begin{aligned} \frac{P_1}{200 \times (200 \times 10^3)} + 2.2 \times 10^{-4} &= \frac{P_2}{300 \times (50 \times 10^3)} + 1.4 \times 10^{-3} \\ 9.166 \times 10^{-8} P_1 &= 1.18 \times 10^{-3} \\ P_1 &= 12873 \text{ N} \quad \& \quad P_2 = -12873 \text{ N} \end{aligned}$$

$$\text{Hence, } \sigma_1 = 64.4 \text{ N/mm}^2 \quad \& \quad \sigma_2 = -42.9 \text{ N/mm}^2$$

Consider temperature drop:

$$\text{Material 1, } \epsilon_1 = \alpha_1 DT = 1.1 \times 10^{-5} \times (-30) = -3.3 \times 10^{-4}$$

$$\text{Material 12 } \epsilon_2 = \alpha_2 DT = 7.0 \times 10^{-5} \times (-30) = -2.1 \times 10^{-3}$$

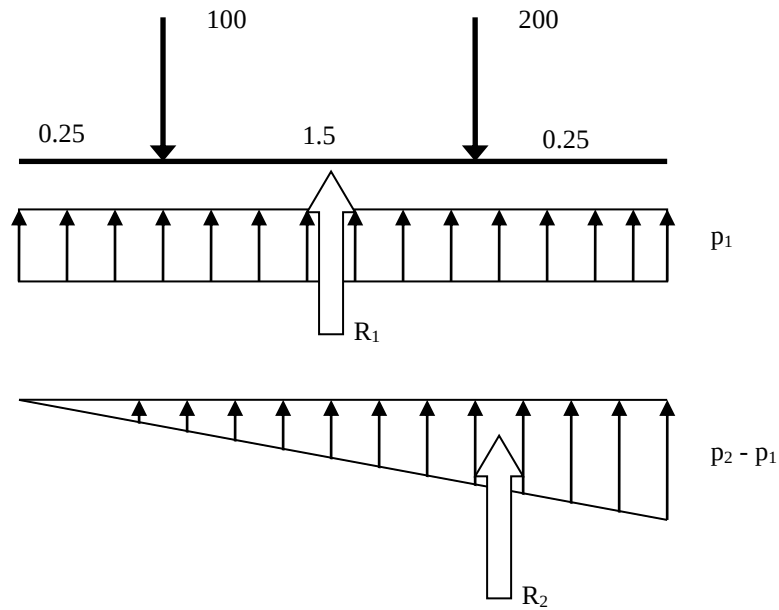
$$\text{Resolving forces: } P_1 + P_2 = 0 \Rightarrow P_2 = -P_1$$

Using compatibility: $\epsilon_1 = \epsilon_2$

$$\begin{aligned} \frac{P_1}{200 \times (200 \times 10^3)} - 3.3 \times 10^{-4} &= \frac{P_2}{300 \times (50 \times 10^3)} - 2.1 \times 10^{-3} \\ 1.77 \times 10^{-3} &= 9.166 \times 10^{-8} P_1 \\ P_1 &= -19310.5 \text{ N} \quad \& \quad P_2 = 19310.5 \text{ N} \end{aligned}$$

$$\text{Hence, } \sigma_1 = -96.6 \text{ N/mm}^2 \quad \& \quad \sigma_2 = 64.4 \text{ N/mm}^2$$

4.



Resolve vertically: $R_1 + R_2 = 100 + 200 = 300$

$$R_1 = (0.25 \times 2.0)p_1 = 0.5p_1$$

$$R_2 = 0.5 \times (0.25 \times 2.0)(p_2 - p_1) = 0.25(p_2 - p_1)$$

$$0.25p_1 + 0.25p_2 = 300$$

$$\therefore p_1 + p_2 = 1200 \quad [1]$$

Take moments about left hand end

$$R_1 \times 1.0 + R_2 \times 4/3 = 100 \times 0.25 + 200 \times 1.75$$

$$3R_1 + 4R_2 = 75 + 1050 = 1125$$

$$0.5p_1 + p_2 = 1125 \quad [2]$$

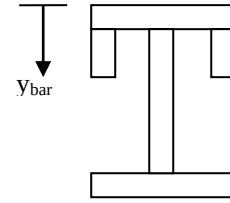
Solve [1] and [2] simultaneously to obtain $p_1 = 150 \text{ kN/m}^2$ and $p_2 = 1050 \text{ kN/m}^2$.

5.

$$\text{Area, } A = (55 \times 5) + 2(15 \times 5) + (70 \times 5) + (55 \times 5) = 1050 \text{ mm}^2$$

$$A y_{\text{bar}} = (55 \times 5 \times 2.5) + 2(15 \times 5 \times 12.5) + (70 \times 5 \times 40) + (55 \times 5 \times 77.5)$$

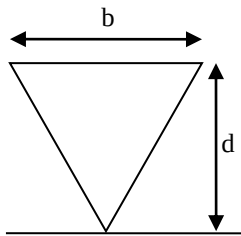
$$y_{\text{bar}} = 37875.0 / 1050 = 36.07 \text{ mm}$$



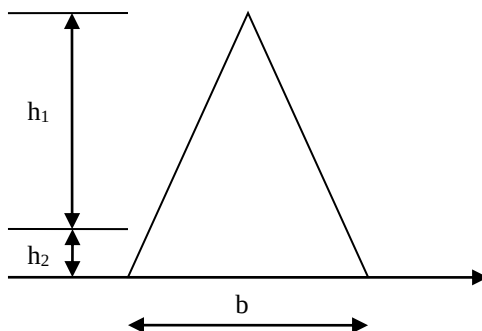
$$\begin{aligned} I &= 55 \times 5^3 / 12 + (55 \times 5)(36.07 - 2.5)^2 + \\ &\quad 2(5 \times 15^3 / 12 + (5 \times 15)(36.07 - 12.5)^2) + \\ &\quad 5 \times 70^3 / 12 + (5 \times 70)(40 - 36.07)^2 + \\ &\quad 55 \times 5^3 / 12 + (55 \times 5)(77.5 - 36.07)^2 = 1017544.6 \end{aligned}$$

$$I = 1.018 \times 10^6 \text{ mm}^4.$$

6.



$$d = \frac{b\sqrt{3}}{2} \quad I_1 = \frac{bd^3}{4} = \frac{b}{4} \left[\frac{b\sqrt{3}}{2} \right]^3 = b^4 \frac{3\sqrt{3}}{32}$$

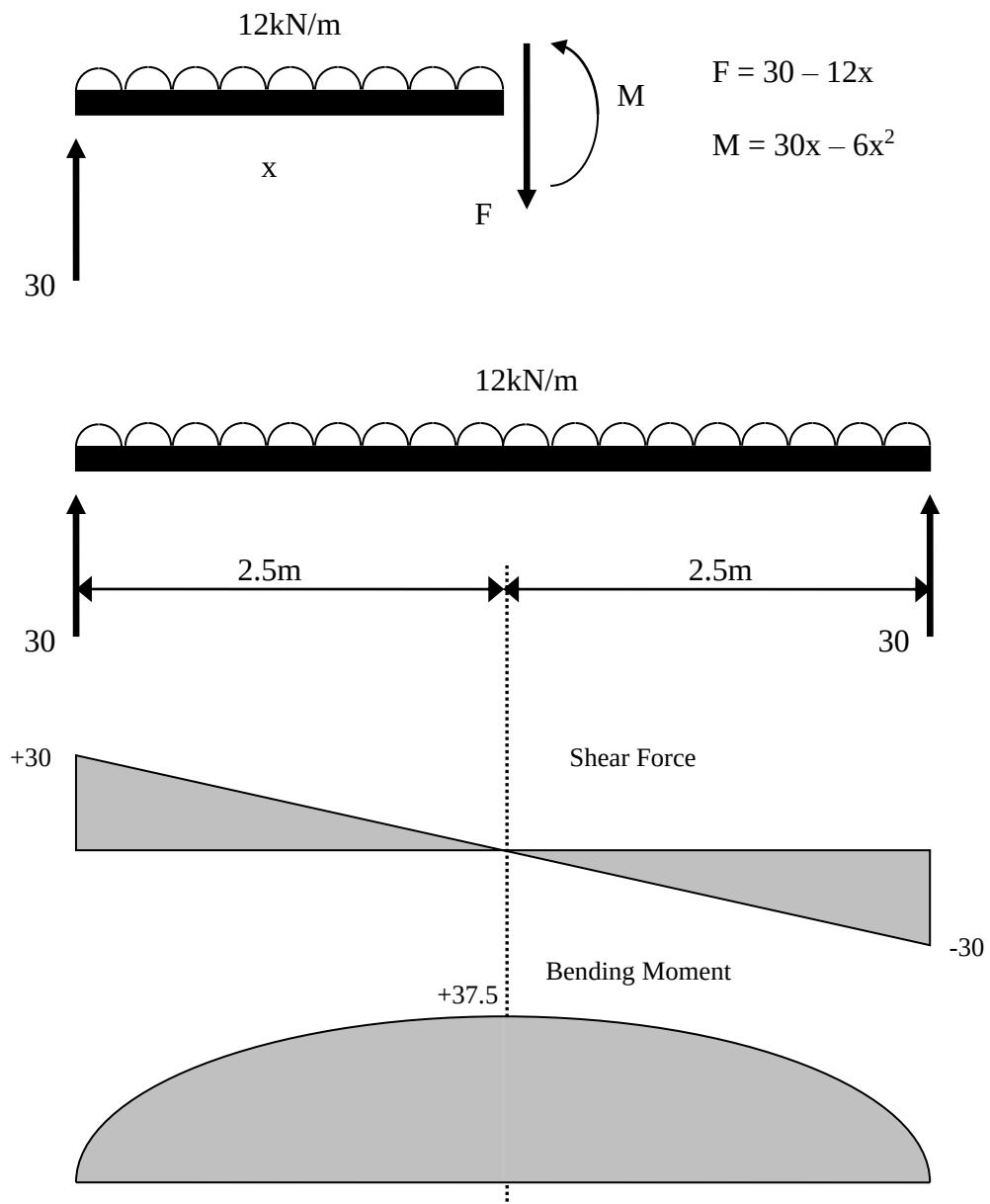


$$h_1 = b / \sqrt{3} \quad h_2 = b / (2\sqrt{3})$$

$$I_2 = \frac{bd^3}{36} + \frac{bd}{2} \left[\frac{d}{3} \right]^2 = \frac{b}{36} \left[\frac{b\sqrt{3}}{2} \right]^2 + \frac{b}{2} \frac{b\sqrt{3}}{2} \left[\frac{b}{2\sqrt{3}} \right]^2 = b^4 \left[\frac{\sqrt{3}}{96} + \frac{\sqrt{3}}{48} \right] = \frac{\sqrt{3}}{32} b^4$$

$$I_{\text{total}} = 2I_1 + 4I_2 = b^4 \left[\frac{3\sqrt{3}}{16} + \frac{\sqrt{3}}{8} \right] = b^4 \sqrt{3} \left[\frac{3}{16} + \frac{1}{8} \right] = \frac{5\sqrt{3}}{16} b^4$$

7.



$$F_{\max} = 30\text{ kN}; M_{\max} = 37.5\text{ kNm}$$